

Research Article

Deep Learning for Histopathological Assessment of Esophageal Adenocarcinoma Precursor Lesions

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ABSTRACT

Histopathological assessment of esophageal biopsies is a key part in the management of patients with Barrett esophagus (BE) but prone to observer variability and reliable diagnostic methods are needed. Artificial intelligence (AI) is emerging as a powerful tool for aided diagnosis but often relies on abstract test and validation sets while real-world behavior is unknown. In this study, we developed a 2-stage AI system for histopathological assessment of BE-related dysplasia using deep learning to enhance the efficiency and accuracy of the pathology workflow. The AI system was developed and trained on 290 whole-slide images (WSIs) that were annotated at glandular and tissue levels. The system was designed to identify individual glands, grade dysplasia, and assign a WSI-level diagnosis. The proposed method was evaluated by comparing the performance of our AI system with that of a large international and heterogeneous group of 55 gastrointestinal pathologists assessing 55 digitized biopsies spanning the complete spectrum of BE-related dysplasia. The AI system correctly graded 76.4% of the WSIs, surpassing the performance of 53 out of the 55 participating pathologists. Furthermore, the receiver-operating characteristic analysis showed that the system's ability to predict the absence (nondysplastic BE) versus the presence of any dysplasia was with an area under the curve of 0.94 and a sensitivity of 0.92 at a specificity of 0.94. These findings demonstrate that this AI system has the potential to assist pathologists in assessment of BE-related dysplasia. The system's outputs could provide a reliable and consistent secondary diagnosis in challenging cases or be used for triaging low-risk nondysplastic biopsies, thereby reducing the workload of pathologists and increasing throughput.

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Introduction

Esophageal adenocarcinoma (EAC) is often diagnosed at a late stage, resulting in a poor prognosis for the patient. Barrett

esophagus (BE) is the only known precursor lesion of EAC. It is characterized by the replacement of squamous epithelium by columnar epithelium in the distal part of the esophagus due to gastroesophageal reflux disease.¹ Patients with BE are at an increased risk of malignant progression, through a gradual process from nondysplastic BE (NDBE) to low-grade dysplasia (LGD), to high-grade dysplasia (HGD) and eventually EAC. Timely diagnosis

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of BE-related dysplasia and early cancer enables endoscopic treatment, resulting in excellent long-term survival rates.² Present guidelines recommend surveillance endoscopies with biopsies for BE patients to detect early neoplasia at a treatable stage.³ Histopathological assessment of esophageal biopsies is a key part in the management of BE patients. A histopathological diagnosis of LGD remains the most reliable predictor of progression to EAC⁴ and is used to determine whether endoscopic surveillance should be continued or intensified, or if endoscopic treatment should be advised.

The assessment of BE-related dysplasia by pathologists is hampered by significant observer variability, and reported progression rates to EAC can vary widely from 1% to 40%.⁵ International guidelines, therefore, mandate review by an expert pathologist.⁴ However, the availability of such expertise is limited, and this shortage is projected to escalate in the coming years as a result of an expected global shortage of pathologists.⁶ Consequently, there is a crucial need to enhance diagnostic methods for assessment of BE biopsies.

Artificial intelligence (AI) has emerged as a powerful tool in various medical disciplines, and deep learning is the most applied AI technology for the analysis of medical images and for computer-aided diagnosis.^{7,8} It has shown its potential in histopathology, ranging from simple applications such as mitosis detection⁹ to more advanced applications such as detection of lymph node metastases in breast cancer¹⁰ or Gleason grading in prostate cancer.¹¹ Compared with studies in advanced cancer, the application of AI models for histopathology assessment of precursor lesions of cancer and especially for the interpretation of BE biopsies has been limited.^{12–14}

In general, studies have shown promise in the interpretation of histopathology data using AI, but most studies lack transparency in the diagnostic process, as they do show the characteristic in situ factors underlying the diagnosis. Furthermore, there is a critical gap in evaluating the accuracy of these AI models, as abstract tests and validation sets are used while real-world behavior is unknown, particularly when compared with both community practice and expert opinions. Such evaluations are crucial for establishing the true potential and effectiveness of AI systems in clinical settings.

Therefore, the objective of this study was to develop an AI system that can automate the assessment of BE biopsies and that is able to visualize grading of dysplasia and to compare its accuracy with assessments made by both community practice and expert pathologists.

The proposed system was designed with 2 distinct stages. In the initial stage, the system generates segmentations at the gland level, aiming to accurately identify and grade individual glands to provide a clearer understanding of the system's final assessments. In the second stage, regional dysplasia scores are derived from these segmentations and used to make a diagnosis at the slide level.

Finally, our proposed method was evaluated by comparing its performance with that of an international and heterogeneous group of 55 gastrointestinal (GI) pathologists assessing 55 digitized biopsies spanning the complete spectrum of BE-related dysplasia.

Materials and Methods

Data Acquisition, Processing, and Annotation

A total of 345 whole-slide images (WSIs) were used for method development and validation, obtained from 2 distinct data sets: an internal data set and the previously described “Bolero” data set¹⁵

Table 1

An overview of the data sets that were used in this study

Data set	Unit	Total	Grade			
			NDBE	IND	LGD	HGD
Internal	WSIs	290	106	0	93	91
	Annotated biopsies	757	464	0	150	143
	Gland annotations	42,273	32,451	0	5139	4683
Bolero	WSIs	55	16	2	22	15

For model development, indefinite cases were excluded in the internal data set. For the Bolero data set, the consensus output of the consensus reference panel is shown based on the hematoxylin and eosin + p53 assessment.

HGD, high-grade dysplasia; IND, indefinite for dysplasia; LGD, low-grade dysplasia; NDBE, nondysplastic Barrett esophagus; WSI, whole-slide image.

(Table 1). The Medical Research Involving Human Subjects Act did not apply to this study. Official approval for this study was therefore waived by the medical ethics review committee (METC W21_042 #21.048).

For the development of the internal data set, hematoxylin and eosin (H&E)-stained WSIs were randomly retrieved from the archives of the Department of Pathology, Amsterdam University Medical Centers, location Amsterdam University Medical Center, Amsterdam, the Netherlands. When the original clinical read made by a GI pathologist was confirmed by a second expert GI pathologist (S.L.M.), cases were used for model development. The final internal data set of 290 WSIs contained a diverse mix of cases from across the diagnostic spectrum of BE (106 NDBE, 93 LGD, and 91 HGD), including referral cases from general hospitals (Table 1). WSIs were digitized using a Philips IntelliSite Ultrafast scanner (Philips Digital Pathology Solutions) at $\times 40$ magnification (0.25 $\mu\text{m}/\text{pixel}$). Each WSI contained 1 to 5 unique biopsies, resulting in a data set containing 757 individual biopsies. A pathologist specialized in BE pathology (S.L.M.) with over 15 years of experience meticulously identified and outlined glands containing NDBE, LGD, or HGD of the initial 100 cases at the pixel level using the open-source annotation tool ASAP (Computational Pathology Group, Radboud University Medical Center).¹⁶ If necessary, ancillary immunohistochemical (IHC) staining for the p53 protein was used to delineate regions of dysplasia. To expedite the annotation process for the remaining cases, a semiautomatic labeling approach (detailed in [Supplementary Methods](#)) was employed. This involved using a U-Net¹⁷ model to automatically delineate the glands, followed by manual correction and classification of each gland as NDBE, LGD, or HGD by S.L.M. For the internal data set, cases diagnosed as indefinite for dysplasia (IND, no definitive diagnosis) were excluded as reproducible pixel-level annotation for this class was not possible. We randomly split the 290 WSIs into training (80%), validation (10%), and internal test sets (10%) with a comparable distribution of dysplasia grades.

To compare the results of the AI system with those of clinical practice, histopathology slides from the Bolero study (the Bolero data set) were used. In the Bolero study, diagnostic concordance rates among pathologists diagnosing BE dysplasia from WSIs were examined. All the 55 BE biopsies assessed in this study, representing a diverse mix of cases across the diagnostic spectrum, were included and underwent the same staining and scanning procedures as the internal data set. A total of 55 participating GI pathologists from 20 countries, from both academic and general hospital settings, individually assessed each case, according to the revised Vienna classification¹⁸ twice: first, only using the H&E WSI, and second, with additional p53 IHC staining, referred to as H&E and H&E + p53 assessments, respectively. To obtain a reference diagnosis for each case, 4 BE pathologists (who were among the 55 participants), further referred to as reference

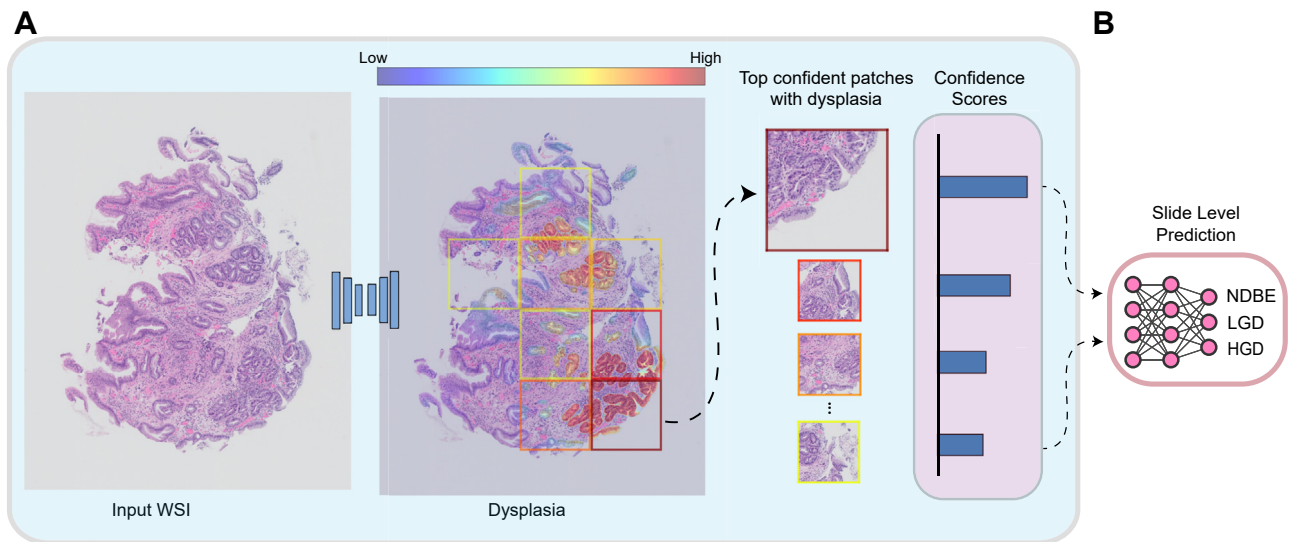


Figure 1.

An overview of the artificial intelligence system for assessment and grading of Barrett esophagus dysplasia. (A) An input whole-slide image is processed by a dysplasia grading network, based on the U-Net architecture. (B) Subsequently, a second network, based on the Set Transformer architecture, takes in confidence scores from highly certain areas of dysplasia for the slide-level prediction into the 3 categories: nondysplastic, low-grade, and high-grade dysplasia. In this example, the dysplasia scoring algorithm detected high levels of dysplasia shown by the dark orange and red colored tissue regions. HGD, high-grade dysplasia; LGD, low-grade dysplasia; NDBE, nondysplastic Barrett esophagus; WSI, whole-slide image.

pathologists, reviewed all cases as a consensus reference panel twice: first, only using the H&E WSI, and second, with additional p53 IHC staining. In cases in which there was no clear consensus, agreement was achieved in a joint panel discussion. The reference pathologists have extensive experience in BE dysplasia assessment and have previously collaborated in multiple intervention studies in which patient outcome has been evaluated prospectively.^{5,19–22} The final consensus output based on the H&E + p53 assessment was considered as the gold standard for this study, resulting in 16 NDBE, 2 IND, 22 LGD, and 15 HGD slides¹⁵ (Table 1).

Major overinterpretation was defined as an NDBE reference diagnosis interpreted as either LGD or HGD, whereas, vice versa, major underinterpretation is an LGD or HGD reference diagnosis underinterpreted as NDBE by the participating pathologists.

Computer Vision System for Assessing Barrett Esophagus–Related Dysplasia

The developed AI system followed a 2-stage approach, as illustrated in Figure 1. In the first stage, a first pass is done over the WSI to identify dysplastic regions and perform dysplasia grade prediction on the pixel level. In the second stage, the pixel-level dysplasia scores obtained from the first stage were used to generate a diagnosis at the slide level. This 2-stage approach was designed to replicate the pathology workflow, wherein individual glands are first examined for signs of dysplasia, and then the findings are aggregated to provide a diagnosis at the slide level, while at the same time attempting to remove any unreliable predictions made at the glandular level.

For the first stage, we utilized the U-Net¹⁷ architecture, and in the second stage, we used the Set Transformer²³ architecture. The U-Net model was trained with H&E-stained image patches of 512×512 pixels at $\times 10$ magnification ($1 \mu\text{m}/\text{pixel}$) to grade dysplasia on a pixel level. This ensured that large parts of the glands were contained within the image patch. We employed a

balanced sampling strategy to sample image patches from annotated regions from the internal training data set, ensuring that patches from all 3 classes (NDBE, LGD, and HGD) were sampled equally.

The Set Transformer model was trained to predict the slide-level diagnosis, given a set of scores derived from the segmented biopsies in the WSI, namely the confidence scores of the N top-ranked patches found in the WSI to contain dysplasia. To extract this set of confidence scores, each WSI was processed as a set of overlapping patches to which the pixel-level dysplasia grading network was applied first. The patches were subsequently ranked based on their confidence in containing dysplasia. The resulting ranked set of confidence scores was used as input for the Set Transformer model and used to predict the slide-level diagnosis.

A more detailed description of the network architectures and training procedures for both stages can be found in [Supplementary Methods](#).

Statistical Analysis

We evaluated the system's performance to grade dysplasia, using accuracy as the primary performance metric, as well as to distinguish any dysplasia (LGD and HGD, including IND) from no dysplasia (NDBE), using the area under the receiver-operating characteristic (ROC) curve as the performance metric. To perform the ROC analysis, we summed the predicted probabilities for LGD and HGD generated by the AI system and used them to establish thresholds. Performance metrics on the internal test set were computed using a 3-tiered grading system (NDBE, LGD, and HGD) because IND cases were not used in training and testing. Additionally, the system's performance was compared with human performance on the Bolero test set, based on the H&E and H&E + p53 assessments. This comparison involved calculating the difference in accuracy of the AI system and the 55 individual pathologists, assigning cases into 1 of 4 diagnostic categories, including the gold standard

Table 2

Cross table comparing the AI system's diagnoses to the consensus-derived reference diagnoses for 29 biopsy cases of the internal test set

Grading dysplasia	AI system			Fraction (%)		
	NDBE	LGD	HGD	Underinterpret	Overinterpret	Correct
Consensus reference panel						
NDBE	7	1	0	/	12.5	87.5
LGD	2	7	1	20	10	70
HGD	0	3	8	27.3	/	72.7
Total				17.2	6.9	75.9
NDBE vs LGD/HGD (optimized threshold tuned for binary classification)	NDBE	Dysplasia (LGD/HGD)				
Consensus reference panel						
No dysplasia	7	1		/	12.5	87.5
Dysplasia	1	20		4.8	/	95.2
Total				3.4	3.4	93.2

Results under heading "Grading dysplasia" are for grading dysplasia (NDBE vs LGD vs HGD) using the default decision threshold of the AI system for grading. Results under heading "NDBE vs LGD/HGD" are for discriminating no dysplasia from dysplasia (LGD or HGD) using the optimal decision threshold determined through ROC analysis for binary classification.

AI, artificial intelligence; HGD, high-grade dysplasia; LGD, low-grade dysplasia; NDBE, nondysplastic Barrett esophagus.

reference diagnosis of IND, as well as the median accuracy of the 51 participating GI pathologists (excluding the 4 reference pathologists).

Results

Development of an Artificial Intelligence System for Assessing Barrett Esophagus–Related Dysplasia

A total of 290 WSIs were used to develop an AI system for grading dysplasia in BE biopsies. After development, the 2 stages of the system were evaluated separately on the internal test set of 29 WSIs.

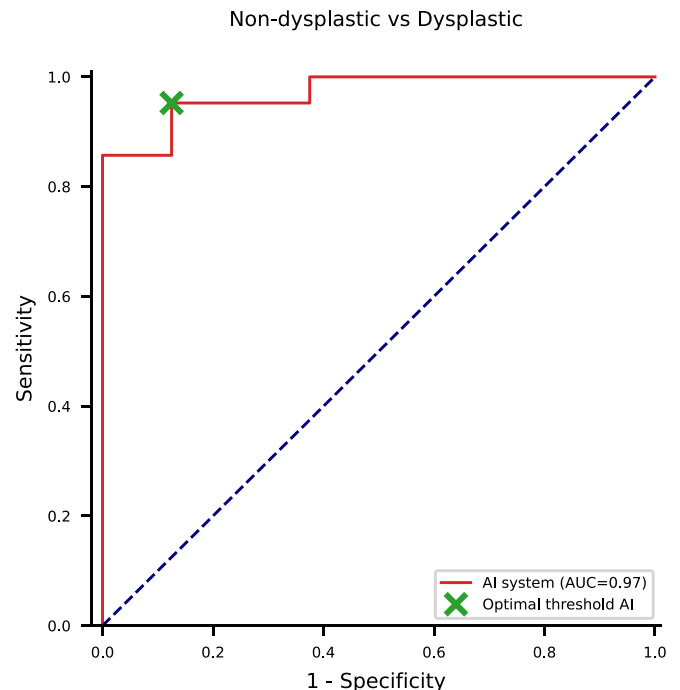
In [Supplementary Figure S1](#), semantic segmentation outputs of the initial stage of the AI system are shown, demonstrating its proficiency in gland segmentation and dysplasia detection. The results reveal that the system can accurately identify dysplastic regions that correspond with the annotations provided by an expert pathologist. For more detailed quantitative outcomes of the gland segmentation stage, please refer to [Supplementary Table S1](#).

On WSI level, the AI system correctly graded 75.9% of the cases of the internal test set as NDBE, LGD, or HGD, when compared with the reference diagnosis. Based on grade, 87.5% of NDBE cases, 70.0% of LGD cases, and 72.7% of HGD cases were correctly identified ("Grading dysplasia"; [Table 2](#)). By using the default decision threshold for grading, major over- or underinterpretation was found in 3 (10.3%) of 29 cases, which was as follows: 1 NDBE case was overdiagnosed as LGD and 2 LGD cases were underdiagnosed as NDBE.

To evaluate the system's accuracy in distinguishing nondysplastic from dysplastic BE, LGD and HGD were grouped. In the ROC analysis, the system achieved an area under the curve (AUC) of 0.97 ([Fig. 2](#)). Moreover, the decision threshold of the AI system could be adjusted to achieve an accuracy of 93.2%, with a sensitivity of 0.95, specificity of 0.88, positive predictive value of 0.95, and negative predictive value of 0.88. By using this decision threshold, 1 case that was initially underinterpreted was now correctly classified as dysplastic by the AI system, lowering the major over- or underinterpretation rate to 2 (6.9%) of 29 ("NDBE vs LGD/HGD"; [Table 2](#)).

Diagnostic Accuracy of an Artificial Intelligence Model Assessing Barrett Esophagus Biopsies Compared With an International Panel of Pathologists

The performance of the developed AI system was compared with that of a group of 55 pathologists who evaluated 55 WSIs of BE-related dysplasia in an international interobserver study.¹⁵ When compared with the consensus reference panel diagnoses

**Figure 2.**

The receiver-operating characteristic curve for artificial intelligence (AI) discrimination of dysplastic versus nondysplastic Barrett esophagus biopsies on the internal test set (n = 29). For the receiver-operating characteristic analysis, the probabilities for low-grade dysplasia and high-grade dysplasia generated by the AI system were summed and used to define thresholds. The optimal decision threshold of the AI, depicted as the green cross, was selected by taking the point closest to the top-left corner. AUC, area under the curve.

Table 3

A cross table comparing the AI system's diagnoses with the consensus-derived reference diagnoses for 55 biopsy cases of the Bolero test set

Consensus reference panel diagnosis based on	AI system				Fraction (%)		
	NDBE	IND	LGD	HGD	Underinterpret	Overinterpret	Correct
H&E assessment							
NDBE	14	0	2	0	/	12.5	87.5
IND	0	0	6	0	0	100	0
LGD	3	0	14	1	16.7	5.5	77.8
HGD	0	0	5	10	33.3	/	66.7
Total					14.5	16.4	69.1
H&E + p53 IHC assessment							
NDBE	14	0	2	0	/	12.5	87.5
IND	0	0	2	0	0	100	0
LGD	3	0	18	1	13.6	4.6	81.8
HGD	0	0	5	10	33.3	/	66.7
Total					14.5	9.1	76.4
NDBE vs IND/LGD/HGD (optimized threshold tuned for binary classification)							
NDBE	15	1			/	6.3	93.7
IND or LGD or HGD	3	36			7.7	/	92.3
Total					5.5	1.8	92.7

Results under the heading “H&E assessment” show grading dysplasia using the Vienna classification compared with the consensus derived from H&E assessment. Results under the heading “H&E + p53 IHC assessment” illustrate dysplasia grading compared with the consensus derived from the H&E + p53 assessment. Results under the heading “NDBE vs IND/LGD/HGD” depict the discrimination between no dysplasia and any dysplasia (IND, LGD, or HGD). Note that here, comparisons with both H&E and H&E + p53 consensus diagnoses yielded similar results. In the first 2 sections of the table (under headings “H&E assessment” and “H&E + p53 IHC assessment”), the AI system used the default decision threshold for grading. In the binary setting in the third section (under the heading “NDBE vs IND/LGD/HGD”), the AI system used the optimal decision threshold determined through ROC analysis for binary classification.

AI, artificial intelligence; H&E, hematoxylin and eosin; HGD, high-grade dysplasia; IHC, immunohistochemical; IND, indefinite for dysplasia; LGD, low-grade dysplasia; NDBE, nondysplastic Barrett esophagus.

based on H&E assessment alone, 69.1% of the cases were correctly graded according to the Vienna classification. Specifically, 14 (87.5%) of 16 NDBE cases, 14 (77.8%) of 18 LGD cases, and 10 (66.7%) of 15 HGD cases were correctly graded (“H&E assessment”; Table 3). All 6 cases diagnosed as IND by the reference consensus panel were classified as LGD by the AI system. When compared with the consensus reference panel diagnosis based on H&E and p53 IHC assessment together, 76.4% of the cases were correctly graded. Specifically, 14 (87.5%) of 16 NDBE cases, 18 (81.8%) of 22 LGD cases, and 10 (66.7%) of 15 HGD cases were correctly graded (“H&E + p53 IHC assessment”; Table 3). Two of the 2 cases classified as IND by the consensus reference panel were diagnosed as LGD by the AI system.

In terms of major errors, the AI system exhibited a major over- or underinterpretation rate of 9.1% of the cases compared with both the H&E and H&E + p53 assessments (“H&E assessment” and “H&E + p53 IHC assessment”; Table 3). There were 2 cases of major overinterpretation (2 of 16 consensus NDBE cases classified as LGD) and 3 cases of major underinterpretation (3 of 22 LGD cases classified as NDBE). Underinterpretation of HGD as NDBE and overinterpretation of NDBE as HGD were not observed.

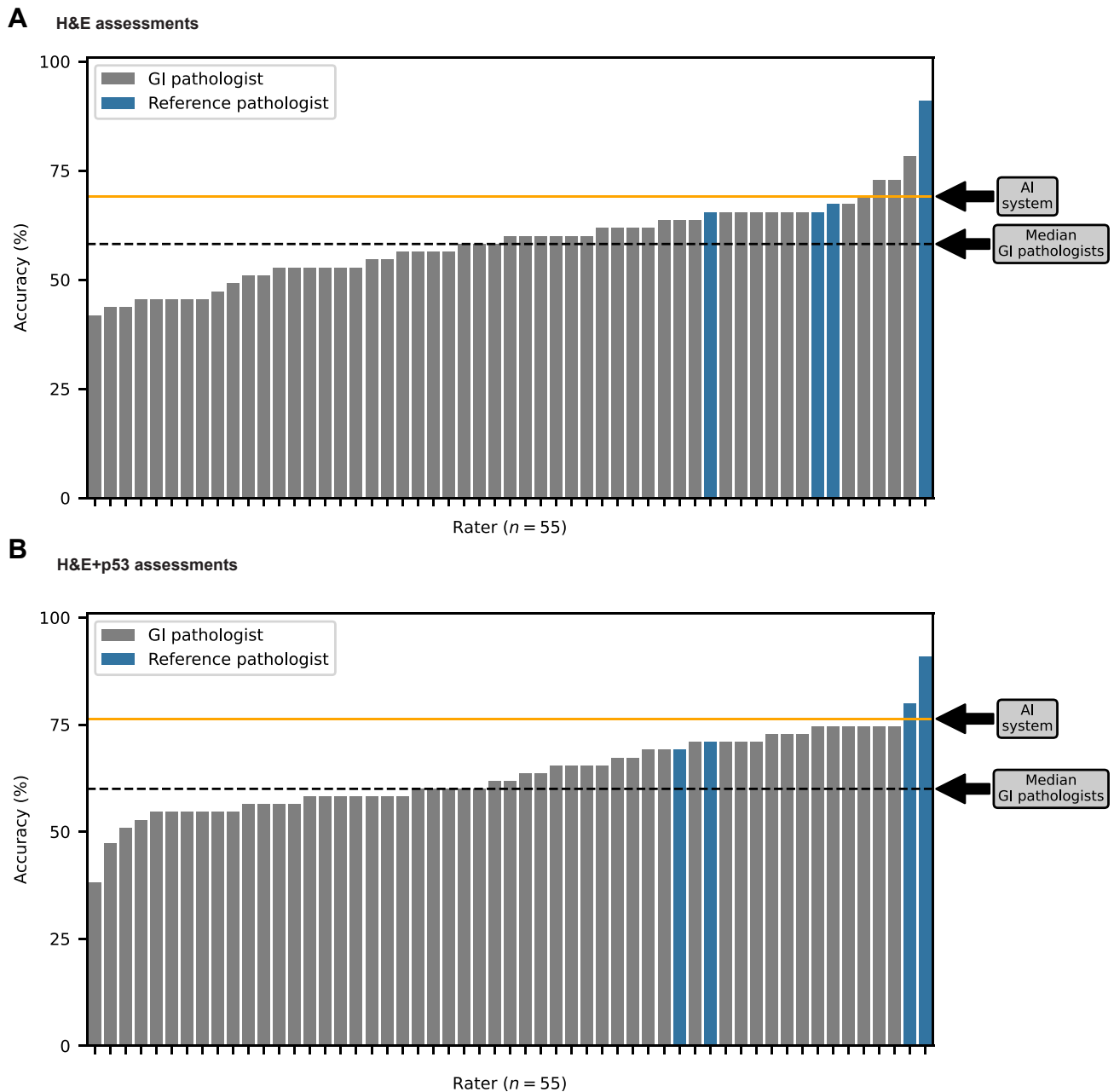
Next, we compared the AI system's performance with that of all 55 pathologists and observed that the system outperformed pathologists in classifying dysplasia. When compared with H&E assessments, the AI system achieved an accuracy of 69.1%, surpassing the median accuracy of 58.2% achieved by the GI pathologists. The AI system outperformed 50 out of the 55 participating pathologists, including 3 out of the 4 reference pathologists (Fig. 3A). When compared with the H&E + p53 assessments, the AI system achieved an accuracy of 76.4%, surpassing the median accuracy of 60.9% achieved by the GI pathologists. The AI system outperformed 53 out of the 55 participating pathologists, including 2 out of the 4 reference pathologists (Fig. 3B).

In terms of distinguishing between the absence of dysplasia (NDBE) and the presence of any dysplasia (IND, LGD, and HGD), identical scores were obtained in the cross tables when comparing the H&E and H&E + p53 diagnoses (both shown in “NDBE vs IND/LGD/HGD”; Table 3). The AI system achieved an AUC of 0.94 in the ROC analysis (in comparison with H&E assessments shown in Fig. 4A and H&E + p53 assessments in Fig. 4B). The analysis revealed that the decision threshold of the AI system can be adjusted to achieve an accuracy of 92.7% (sensitivity: 0.92; specificity: 0.94; positive predictive value: 0.97; and negative predictive value: 0.83). Using this decision threshold, 1 additional case was correctly reclassified from LGD to NDBE by the AI system, lowering the major over- or underdiagnosis rate to 4 (7.3%) of 55 (“NDBE vs IND/LGD/HGD”; Table 3).

In Figure 5, we illustrate a case study involving a WSI that was accurately diagnosed by the AI system but underdiagnosed by a substantial number of the participating pathologists. This example showcases the AI system's precision in identifying dysplasia regions within the WSI. The assigned reference diagnosis for this WSI was LGD, with a consensus among 3 out of 4 reference pathologists concurring on this assessment, whereas 1 pathologist diagnosed it as NDBE. Among the remaining 51 pathologists, their diagnoses comprised 21 cases of NDBE, 14 cases of IND, 11 cases of LGD, and 5 cases of HGD. The AI system accurately diagnosed this WSI as LGD, with probability estimates of 24% for NDBE, 54% for LGD, and 22% for HGD. Supplementary Figure S2 illustrates an example highlighting the suboptimal behavior of the AI system compared with that of the participating pathologists.

Discussion

In this study, we describe the development of a 2-stage AI system for the assessment of BE-related dysplasia in esophageal

**Figure 3.**

Diagnostic accuracy comparison for Barrett esophagus–related dysplasia grading on the Bolero test set (n = 55). The figures show the accuracy of the artificial intelligence system compared with that of 55 pathologists in grading Barrett esophagus–related dysplasia according to the Vienna classification (nondysplastic Barrett esophagus, indefinite for dysplasia, low-grade dysplasia, and high-grade dysplasia). The accuracy percentages reflect agreement with the consensus reference panel. (A) Data from assessments using hematoxylin and eosin (H&E) stains. (B) Results from H&E coupled with p53 immunohistochemistry staining. Each pathologist's accuracy is displayed. The artificial intelligence system only had access to the H&E slide as the input for diagnosis prediction. AI, artificial intelligence; GI, gastrointestinal.

biopsies. The initial stage involves the precise assessment of dysplasia at the pixel level, followed by a subsequent diagnosis at the WSI level in the second stage. The AI system shows high sensitivity and specificity, and when accuracy is compared with that of a large international panel of pathologists, it outperforms 53 out of the 55 participating pathologists. These results provide valuable insights into the potential and effectiveness of computer diagnosis in a clinical setting.

Three prior studies have addressed the automated evaluation of BE biopsies through the application of deep learning

techniques. In one study,¹² an attention-based deep neural network was developed to diagnose a broad range of different tissue types including normal squamous epithelium, nondysplastic BE, dysplastic BE, and adenocarcinoma. Sensitivities in the range of 0.21 to 0.77 were reported, alongside an overall accuracy of 83% for 379 single biopsy images when compared against diagnoses made by 2 expert pathologists. However, this study did not encompass the histologic discrimination between LGD and HGD, an important clinical challenge in the management of BE patients. In addition, in this study the diagnosis was solely made at

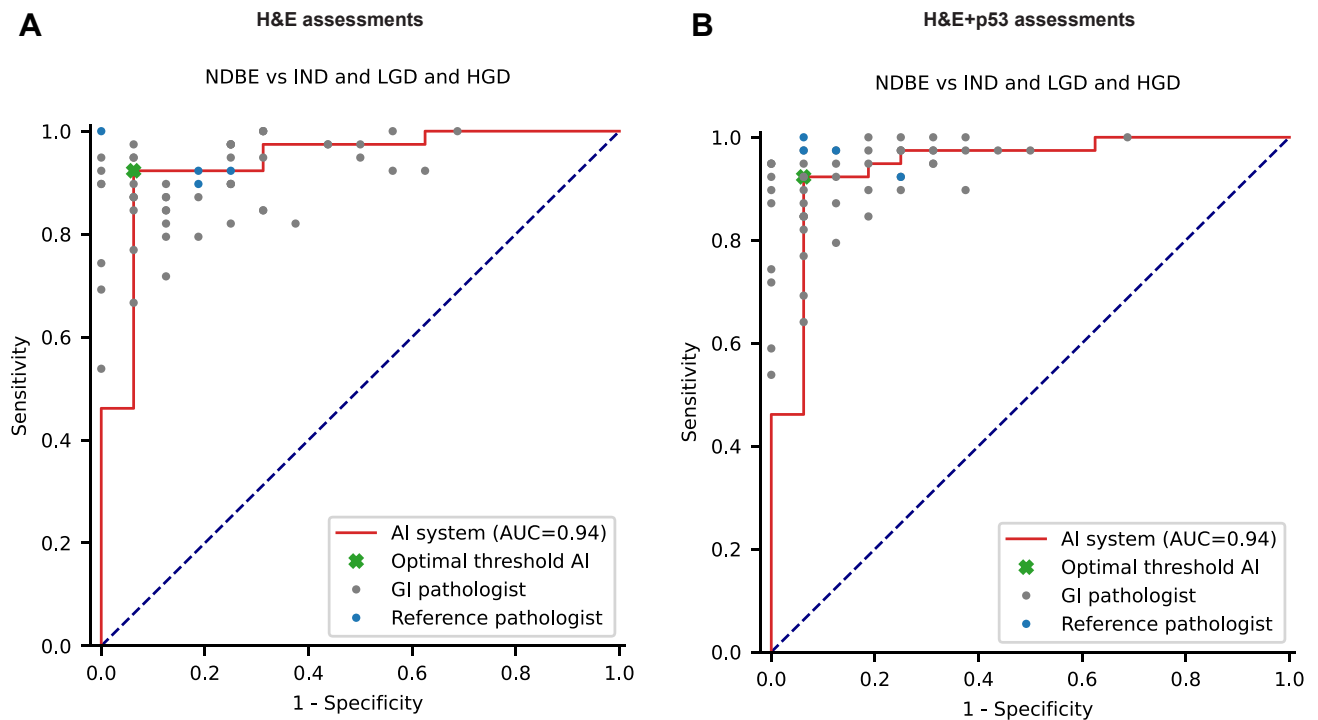


Figure 4.

Receiver-operating characteristic (ROC) analysis of the artificial intelligence (AI) system's diagnostic accuracy on the Bolero test set ($n = 55$). This figure illustrates the AI system's performance in distinguishing between no dysplasia and any level of dysplasia (IND, LGD, and HGD). (A) The ROC curve for assessments using H&E stains. (B) The ROC curve for assessments using H&E with p53 staining. For this analysis, predicted probabilities for LGD and HGD were combined to establish thresholds. The curves display the AI's ROC alongside individual pathologists' operating points, marked by gray dots for gastrointestinal pathologists and blue for reference pathologists. The optimal AI threshold, indicated by a green cross, represents the point nearest to the top-left corner. AUC, area under the curve; GI, gastrointestinal; H&E, hematoxylin and eosin; HGD, high-grade dysplasia; IND, indefinite for dysplasia; LGD, low-grade dysplasia; NDBE, nondysplastic Barrett esophagus.

a single biopsy level, and not using WSIs with multiple biopsies, each contributing to the comprehensive final diagnosis.

In a second study,¹³ a combination of histologic information from H&E-stained slides and spatially detailed molecular data from mass spectrometry imaging was employed. The classification model on H&E-stained slides was based on a Convolutional Block Attention Module²⁴ with ResNet50²⁵ as the backbone as proposed by the previously mentioned work of Tomita et al¹² with minor modifications. They achieved an AUC of 0.85 in predicting dysplasia grades (NDBE, LGD, and HGD) on tile level. Although this result within the domain of automated pathology assessment exhibits potential, it is important to acknowledge that the transition toward practical clinical application, particularly through a tile-level analysis, remains a substantial distance away.

In another study,¹⁴ an object detection framework approach was introduced along with a distinct classifier to distinguish nondysplasia from LGD and HGD at slide level. This approach performed best at both ends of the spectrum, with reported sensitivities for NDBE of 0.94, LGD of 0.82, and HGD of 0.95. An overall accuracy of 88.6% was reached across the assessment of 70 WSIs from a single institute, assessed against reference diagnoses provided by a relatively small panel of 2 expert GI pathologists. Comparisons were made in a 3-tiered system as cases with a diagnosis of IND, routinely used in daily diagnostic workup of BE biopsies, were not included in this study.

Our study shows several strengths. First, to address the significant interrater variability in BE dysplasia grading, in this study, a strong consensus reference standard of 4 expert pathologists was used, which, to our knowledge, is one of the largest in AI histopathology-related work. Second, the diagnostic accuracy of

our AI model was compared against a large and international group of 55 pathologists, both from private, community practice and academic settings, and thereby provides a more complete perspective into the potential and effectiveness of computer-assisted diagnosis in a clinical setting. Third, the AI system visualizes the specific region of the BE biopsy the diagnosis is based on. This approach allows transparency for computer-aided diagnoses and enables pathologists to examine the regions of the biopsy with the assigned grades and evaluate the accuracy and added values of the computer diagnosis.

In our 2-stage approach, the AI system was first trained using diagnostic segmentations at the microscopic level for NDBE, LGD, and HGD. However, due to the impossibility of specifying uncertainty at the glandular level (corresponding with the diagnosis of IND), predicting the IND category at glandular level was not included. In the second stage, the diagnostic AI system was evaluated at WSI level, assessing the complete diagnostic spectrum of BE-related dysplasia, including IND; all IND cases were classified as LGD.

Before integration of our system into clinical practice, some limitations must be addressed. Both the internal data set and the Bolero data set consisted of roughly equal amounts of NDBE, LGD, and HGD cases. This distribution is skewed toward the dysplastic spectrum, deviating from the typical distribution of diagnostic biopsy cases encountered in routine clinical practice, in which NDBE cases dominate. Therefore, the evaluation sets fail to reflect the full spectrum of cases encountered in real-world scenarios.

Although 25% of cases used originated from multiple general hospitals, the case mix for the development of the AI system predominantly originated from a single center, and the WSIs were

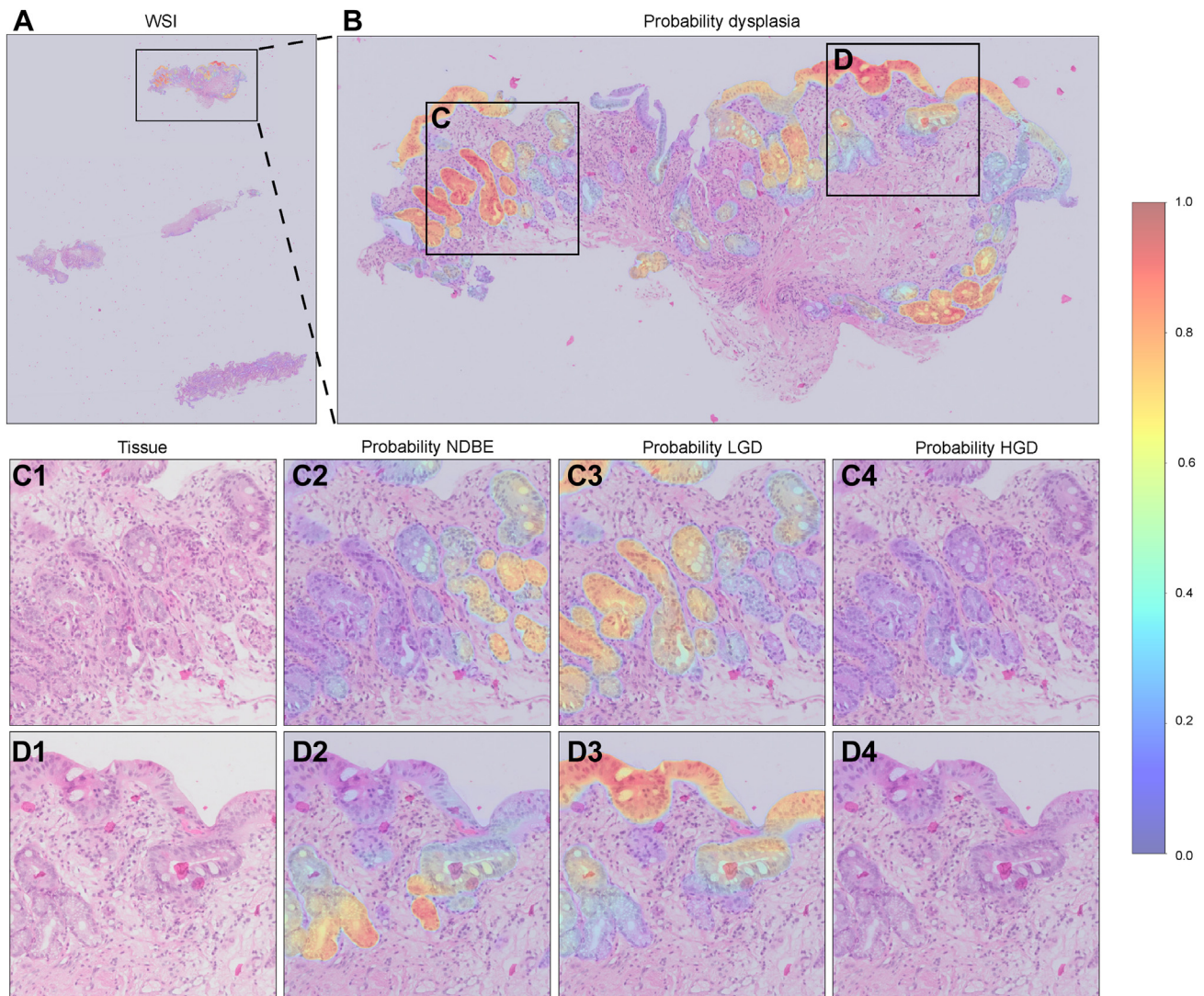


Figure 5.

Example of a case accurately diagnosed by the artificial intelligence (AI) system, but underdiagnosed by multiple pathologists. (A) An example of a whole-slide image (WSI), for which the reference diagnosis was low-grade dysplasia (LGD), with 3 out of 4 reference pathologists agreeing on LGD, and 1 reference pathologist diagnosing the WSI as nondysplastic Barrett esophagus (NDBE). Among the remaining 51 pathologists, 21 diagnosed as NDBE, 14 were indefinite for dysplasia, 11 diagnosed as LGD, and 5 diagnosed as high-grade dysplasia (HGD). The AI system accurately diagnosed this WSI as LGD, with probabilities of NDBE being 24%, LGD being 54%, and HGD being 22%. (B) The biopsy containing dysplasia, accompanied by a heat map showing the AI system's predicted probability of dysplasia. We have highlighted 2 regions, labeled as C and D, to demonstrate the AI system's proficiency in gland segmentation and dysplasia detection. In subpanels c1 and d1, the tissue is displayed without the heat map. In subpanels c2 and d2, c3 and d3, and c4 and d4, the probabilities of NDBE, LGD, and HGD are depicted, respectively, for both the regions.

digitized using a single slide scanner. This particular limitation is also shared by the 3 previously mentioned studies. This approach may introduce biases and limit the generalizability of the results. To enhance the robustness of the system, future work should involve external validation with more extensive multicenter data sets and multiple types of slide scanners. This approach would help validate the system's performance across different settings and strengthen its applicability in a broader range of clinical scenarios.

Considering the current limited availability of expertise for assessing BE biopsies, there is a clear clinical advantage in reducing the workload for pathologists with computer-assisted diagnosis. Our developed system holds promising potential as an auxiliary tool for pathologists, offering rapid and reliable support in dysplasia grading or biopsy screening. On the Bolero test set, our system correctly classified 76.4% of the cases according to

the standard Vienna classification system of NDBE, IND, LGD, and HGD. It was able to predict the presence of any dysplasia (IND, LGD, and HGD) with an AUC of 0.94, a sensitivity of 0.92, and a specificity of 0.94, outperforming the large majority of participating pathologists. In this analysis, we grouped IND with LGD and HGD, as a pathology diagnosis of IND initially leads to a comparable frequency of endoscopic surveillance, as for LGD.

The AI system classified all 6 cases that the reference panel had labeled as IND based on the H&E assessment alone, as LGD. As this category suffers from severe interobserver variation, the AI could not be objectively trained to reproducibly recognize morphologic changes consistent with IND. However, when the reference panel pathologists reassessed H&E WSIs together with p53 IHC, they reclassified 4 of 6 cases from IND as LGD, aligning with the AI system's WSI classification of LGD based on the H&E input alone. This indicates that the AI system was able to accurately classify 4

out of the 6 IND case diagnoses by the reference panel as LGD, a task the experts themselves were unable to perform without additional p53 IHC staining.

Addressing the AI system's limited capacity to express uncertainty corresponding with the diagnosis of IND could, in future work, be achieved by expanded training data of the second stage of the system with IND cases. It is anticipated, however, that the AI system would also face difficulties in classifying IND samples accurately and might require a substantial sample size to be able to diagnose IND at acceptable levels. As IND is a challenging category with very low agreement among pathologists, methods that offer informative probabilistic estimates of diagnoses rendered thereby flagging the biopsies that need special attention by the pathologist might be of more use. To improve the system's discriminatory power, automated assessment of IHC staining for the p53 protein could be potentially incorporated.

The AI system misclassified 5 out of 15 HGD cases as LGD. Although the diagnosis of HGD is usually straightforward, in 4 of these 15 HGD cases, no majority diagnosis was reached among the 4 reference pathologists during individual digital slide reviews. In these 4 cases, a consensus HGD was established only through panel discussions, emphasizing the subtleness of these cases. Interestingly, 3 of these 4 cases were classified as LGD by the AI system. A fourth HGD case diagnosed as LGD by the AI system was diagnosed by 3 of the 4 reference pathologists as HGD and by 1 reference pathologist as LGD. These findings suggest that diagnosing HGD can be difficult even for experienced pathologists, and that the AI system encountered challenges in accurately identifying more subtle HGD cases.

The AI system demonstrated major misinterpretations in 4 (7.3%) of 55 cases using the optimized decision threshold to discriminate between dysplasia and no dysplasia. In comparison, the participating pathologists exhibited an average major misinterpretation rate of 8.8% when assessing H&E WSIs alone and an average of 8.3% when assessing both H&E and ancillary p53 IHC. Although these results are comparable, direct comparison is flawed as the AI system did not classify cases as IND, in contrast to the participating pathologists in up to 21 (38.2%) of 55 cases (mean, 11.2%). A high percentage of IND may keep major under- and over-interpretation rates in diagnostic practice low, but it is of limited help for the gastroenterologist and patients in proper risk stratification. Furthermore, the AI system did not overinterpret NDBE as HGD, as opposed to 1.1% of NDBE cases by the participating pathologists. The system also did not underinterpret HGD as NDBE, as opposed to 2.8% of HGD cases by the participating pathologists with a result of falsely dismissing patients with HGD from endoscopic follow-up.

Ancillary p53 is the only proven biomarker in the assessment of BE biopsies,^{26,27} and aberrant p53 IHC staining can visualize regions of interest that are in line with the morphologic diagnosis of dysplasia in part of BE cases. Our study suggests potential of the AI system in the assessment of H&E WSIs alone, resulting in a lower rate of IND diagnoses in cases in which the majority of (expert) pathologists need both H&E and p53 IHC to render a definite diagnosis. Although the AI system outperforms many pathologists in this study, it is premature to be deployed in settings outside of controlled experiments. Implementation of a combined approach involving p53 IHC and the AI system may already aid in the diagnostic process. Morphologic feedback of regions that warrant further examination highlighted by both methods can affirm the diagnosis and hold extra didactic potential for pathologists on assessing BE biopsies. As regions of interest

visualized by the AI system can be achieved without the need of performing extra p53 IHC tests on parallel slides, the potential hazard of cutting out the specific aberrant region of the tissue block can be eliminated. This can be extra useful in cases in which the aberrant focus is limited and p53 is performed on parallel slides, when access to the tissue blocks is difficult, and in remote (digital) consultation cases. Furthermore, in cases of BE-related dysplasia without aberrant p53 IHC staining patterns, AI system predictions may offer aid in diagnosis. The ultimate goal is to develop an algorithm that performs at the level of expert pathologists and can operate effectively without the need for additional p53 staining.

In conclusion, in this study, we successfully developed a specialized AI system for the automated assessment of BE biopsies. The AI system demonstrated high accuracy, surpassing the performance of 53 out of 55 pathologists when evaluated against a large international panel. This accomplishment highlights the significant potential of our developed system as a valuable tool for pathologists, providing fast and reliable support in dysplasia grading and biopsy screening. However, further investigations are necessary to optimize the system's performance and determine its practical implementation in real-world clinical settings.

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Author Contributions

S.L.M. and O.J.B. were responsible for the conceptualization and design of the study. The development of the methodology was carried out by M.B., B.C., and O.J.B. M.B. wrote the manuscript, for which S.L.M. and O.J.B. provided reviews and revisions. S.L.M. and O.J.B. were involved in the acquisition of the datasets in this study. M.J.W. and M.J. took charge of all matters pertaining to the previously conducted Bolero study. S.L.M. contributed manual annotations crucial for model development. S.L.M., M.J., and M.J.W. provided pathology expertise during the project. M.B. took charge of the analysis, data interpretation, and statistical procedures. E.J.B. and C.I.S. provided essential support in the technical and artificial intelligence aspects during the development of the methodology and the writing of the manuscript. All authors read and approved the final manuscript.

Data Availability

Although the data sets used in this project are not made publicly available, access to the data sets used in this study or details of its acquisition can be made available from the corresponding author on reasonable request.

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Declaration of Competing Interest

All authors declare no competing interests.

Ethics Approval and Consent to Participate

The Medical Research Involving Human Subjects Act did not apply to this study. Official approval for this study was therefore waived by the medical ethics review committee (METC W21_042 #21.048).

Supplementary Material

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